Vladimir Sosnilo

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date of birth: 21.08.1994

Research interests

- Category theory and ∞-category theory.
- Algebraic K-theory, stable ∞-categories, noncommutative motives.
- Algebraic geometry, motivic homotopy theory, descent conditions for cohomology theories.

Employment

2025-now **RIKEN iTHEMS**, Wako, Japan
Research scientist

2022-2025 **University of Regensburg**, Regensburg, Germany
Research assistant

2018-2022 **Laboratory of Modern Algebra and Applications**, *Department of Mathematics & Computer Science of SPbU*, St. Petersburg, Russia
Research Engineer

2013-2018 **Chebyshev Laboratory**, *Mathematics & Mechanics Faculty of SPbU*, St. Petersburg, Russia
Research Engineer

Non-academic employment

2011-2012 **LoyaltyPlant company**, St. Petersburg, Russia
Software Developer

Education

2018-2022 **Phd in Mathematics**, *Higher School of Economics*, Moscow, Russia
Advisor: Ivan Panin

2016-2017 **Master of Mathematics**, *University of Duisburg-Essen*, Essen, Germany
Advisor: Marc Levine

2011-2015 **Bachelor of Mathematics**, *St. Petersburg State University*, St. Petersburg, Russia
Advisor: Mikhail Bondarko

Publications

1 Maxime Ramzi, Vladimir Sosnilo, Christoph Winges. Every motive is the motive of a stable ∞-category. *Accepted in Journal of the European Mathematical Society*. arXiv:2503.11338

2 Ishan Levy, Vladimir Sosnilo. c-structures and trace methods beyond connective rings. *Accepted in Duke Mathematical Journal*. arXiv:2509.14774

3 Andrei Druzhinin, Vladimir Sosnilo. Zariski-local framed $\mathbb{A}^1$-homotopy theory. *To appear in Algebraic & Geometric Topology*. arXiv:2108.08257

4 Vladimir Sosnilo. $\mathbb{A}^1$-invariance of localizing invariants. *Annals of K-theory*, 9 (2024). arXiv:2211.05602

5 Tom Bachmann, Adeel A. Khan, Charanya Ravi, Vladimir Sosnilo. Categorical Milnor squares and K-theory of algebraic stacks. *Selecta Mathematica*, 28 (2022). arXiv:2011.04355

6 Elden Elmanto, Vladimir Sosnilo. On nilpotent extensions of ∞-categories and the cyclotomic trace. *International Mathematics Research Notices*, 21 (2022). arXiv:2010.09155

7 Mikhail Bondarko, Vladimir Sosnilo. On Chow-weight homology of geometric motives. *Transactions of the AMS*, 375 (2022). arXiv:1411.6354

8 Vladimir Sosnilo. Regularity of spectral stacks and discreteness of weight-hearts. *Quarterly Journal of Mathematics*, 73 (2022). arXiv:1901.02431

9 Elden Elmanto, Marc Hoyois, Adeel A. Khan, Vladimir Sosnilo, Maria Yakerson. Motivic infinite loop spaces. *Cambridge Journal of Mathematics*, 9 (2021). arXiv:1711.05248

10 Tom Bachmann, Elden Elmanto, Marc Hoyois, Adeel A. Khan, Vladimir Sosnilo, Maria Yakerson. On the infinite loop spaces of algebraic cobordism and the motivic sphere. *Épijournal de Géométrie Algébrique*, 5 (2021). arXiv:1911.02262

11 Elden Elmanto, Marc Hoyois, Adeel A. Khan, Vladimir Sosnilo, Maria Yakerson. Modules over algebraic cobordism. *Forum of Mathematics, Pi*, 8 (2020). arXiv:1908.02162

12 Elden Elmanto, Marc Hoyois, Adeel A. Khan, Vladimir Sosnilo, Maria Yakerson. Framed transfers and motivic fundamental classes. *Journal of Topology*, 13 (2020), 460-500. arXiv:1809.10666

13 Vladimir Sosnilo. Theorem of the heart in negative K-theory for weight structures. *Documenta Mathematica*, 24 (2019), 2137-2158. arXiv:1705.07995

14 Mikhail Bondarko, Vladimir Sosnilo. On purely generated α-smashing weight structures and weight-exact localizations. *Journal of Algebra*, 535 (2019), 407-455. arXiv:1712.00850

15 Sergei O. Ivanov, Roman Mikhailov, Vladimir Sosnilo. Higher colimits, derived functors and homology. *Sbornik Mathematics*, 210 (2019), 1222-1258. arXiv:1805.07754

16 Mikhail Bondarko, Vladimir Sosnilo. On the weight lifting property for localizations of triangulated categories. *Lobachevskii Journal of Mathematics*, 39 (2018), 970–984. arXiv:1510.03403

17 Mikhail Bondarko, Vladimir Sosnilo. On constructing weight structures and extending them to idempotent extensions. *Homology, Homotopy and Applications*, 20 (2018), pp.37–57. arXiv:1605.08372

18 Mikhail Bondarko, Vladimir Sosnilo. Non-commutative localizations of additive categories and weight structures; applications to birational motives. *Journal of the Institute of Mathematics of Jussieu*, 17 (2018), 785-821. arXiv:1304.6059

19 Mikhail Bondarko, Vladimir Sosnilo. A Nullstellensatz for triangulated categories. *Algebra and Analysis*, 27 (2015), 41-56. arXiv:1508.04427

Preprints

1 Xiao-Wu Chen, Sergei O. Ivanov, Jian Liu, Xue-Song Lu, Vladimir Sosnilo, Chencheng Zhang. Pretriangulated categories failing (TR4). arXiv:2608.09777

2 Ishan Levy, Jiacheng Liang, Vladimir Sosnilo. Dualizable additive categories. arXiv:2608.04898

3 Elden Elmanto, Dmitry Kubrak, Vladimir Sosnilo. On filtered algebraic K-theory of stacks I: characteristic zero. arXiv:2503.09928

4 Maxime Ramzi, Vladimir Sosnilo, Christoph Winges. Every spectrum is the K-theory of a stable ∞-category. arXiv:2401.06510

PhD thesis

Weighted methods in noncommutative geometry. *Defended at the HSE in Moscow.* pdf

Invited Talks & Conferences

Sep, 2026 **c-categories**
CMI Workshop "Telescopic Homotopy Theory and Algebraic K-theory", Oxford University

May, 2026 **c-categories**
Nagoya University

Apr, 2026 **Jouanolou's trick in noncommutative geometry**
Tohoku University Algebra Seminar, Tohoku University

Feb, 2026 **Motives as a localization**
Conference "Motives in Tokyo" on the occasion of Thomas Geisser's 60th Birthday, Tokyo University

Nov, 2025 **Negative K-theory of t-categories**
MS Seminar, KAVLI IPMU

Oct, 2025 **Dualizable additive categories**
Algebraic Geometry Seminar, RIMS

Aug, 2025 **The stacky view on the Grayson construction**
Oberwolfach workshop "Algebraic K-theory", Mathematisches Forschungsinstitut Oberwolfach

Jan, 2025 **Atiyah-Segal completion theorem for singular varieties**
Algebraic Geometry and Number Theory Seminar, ISTA

Nov, 2024 **Resolutions of c-categories**, link
Online Seminar "New Directions in Group Theory and Triangulated Categories"

Jan, 2024 **A counterexample to the theorem of the heart**
BIMSA Topology Seminar

Jun, 2024 **K-theory of nonconnective ring spectra**
International Workshop on Algebraic Topology, Hangzhou

May, 2024 **A counterexample to the nonconnective theorem of the heart**
EPFL topology seminar

Apr, 2024 **Theorems of the heart in algebraic K-theory and their applications in geometry**
lecture series at Graduate School of Mathematical Sciences of Tokyo University

Mar, 2024 **Every noncommutative motive is the motive of a stable ∞-category**
Conference "Motives in Mainz", Mainz University

Jan, 2024 **Trace methods and the Atiyah-Segal completion theorem**
Bangalore Centre of the Indian Statistical Institute

Jun, 2023 **Pro-excision for K-theory of algebraic stacks**
Münster University

May, 2023 **Introduction to weight structures**
Institut de Mathématiques de Jussieu-Paris Rive Gauche

Feb, 2023 **Weighted $\mathbb{A}^1$-invariance and the Atiyah-Segal completion theorem**
Conference "Motives in Tokyo", Graduate School of Mathematical Sciences of Tokyo University

Sep, 2022 **Motives without $\mathbb{A}^1$-invariance**
lecture series at the Summer school "Motives in Ratisbona", Regensburg University

Mar, 2022 **Weight structures and applications I: structural results**
Leiden University

Jan, 2021 **On nilpotent extensions of ∞-categories and the cyclotomic trace**
AG seminar, Regensburg University, online

Oct, 2020 **Comparing Nisnevich descent, Milnor excision, and the pro-cdh excision**
AG seminar, St. Petersburg State University, online

Sep, 2020 **Excision for algebraic K-theory with respect to categorical Milnor squares**
Seminar on A1-topology, motives and K-theory, St. Petersburg State University, online

Jun, 2020 **Pro-excision for stacks**, link
Conference "Motives and What Not", online

Dec, 2019 **Milnor squares and the Weibel's conjecture**
Seminar on A1-topology, motives and K-theory, St. Petersburg State University

Nov, 2019 **Regularity of spectral stacks, their algebraic K-theory, and weight structures**, link
AG seminar, Regensburg University

May, 2018 **K-theory of the category of Voevodsky motives**
St. Petersburg algebraic geometry symposium for young mathematicians, St. Petersburg State University

Dec, 2018 **Weight structures on stable infinity-categories**
Algebra/Topology seminar, University of Copenhagen

Nov, 2017 **Comparing different framed transfers**
Seminar on A1-topology, motives and K-theory, St. Petersburg State University

Jun, 2017 **Theorem of the heart for weight structures**
AG seminar, Regensburg University

Sep, 2016 **Non-commutative localizations of additive categories and weight structures**
Harish-Chandra Research Institute

Apr, 2015 **Noncommutative localizations of additive categories and weight structures**
Faddeev seminar, St. Petersburg Department of Steklov Institute

Awards & Grants

2026–2030 **JSPS KAKENHI Grant-in-Aid for Early-Career Scientists**, *awarded for the project "Noncommutative Descent and Trace Methods"*

2013, 2017 **Rokhlin grant**, *awarded by St. Petersburg Department of Steklov Institute*

Teaching experience

Winter, 2021 **Sheaves and the continuum hypothesis**, *lecture series for the Winter School in Mathematics and Theoretical Computer Science*, link
online

Fall, 2020 **Homotopy type theory**, *Seminar jointly organized with Valery Isaev*, recordings (in Russian)
online

Spring, 2020 **Higher category theory**, *Seminar jointly organized with Andrei Lavrenov*
online

Fall, 2019 **Type theory**, *Seminar jointly organized with Dmitry Shtukenberg*
St. Petersburg State University

Fall, 2015 **Algebraic topology and vector bundles**, *joint with Alexey Ananyevskiy*, lecture series for sophomore undergraduate students
St. Petersburg State University

Fall, 2014 **Linear algebra**, problem sessions for sophomore undergraduate students
St. Petersburg State University

Fall, 2013 **Introduction to commutative algebra**, lecture series for high school students
Laboratory of Continuous Mathematical Education

Summer, 2012 **Combinatorics**, lecture series for high school students
Laboratory of Continuous Mathematical Education

Supervising students

2024-2025 Tianjian Tan: supervising master's thesis on comparing motivic homotopy theories for Sorbonne Université

2022-2024 Nikita Golub: cosupervising master's thesis on quantum field theories and geometry of stacks for SPbU University

2015-2016 Gleb Novikov: a high school student of "Laboratory of Continuous Mathematical Education" completed a research project under my supervision and presented it at the International Science and Engineering Fair in Pittsburgh, Pennsylvania, USA

Service to the community

Referee reports for various mathematical journals including Alg. & Geom. Topology, IMRN, JIMJ, Annals of K-theory, Compt. Rend. Math.

Quick opinions for various mathematical journals including Duke Mathematical Journal, Invent. Math.

Languages

English: fluent
German: beginner
Japanese: intermediate
Russian: native